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An 'other-race effect' for categorizing faces by sex

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Abstract. It is well-known that people recognize faces of their own race more accurately than faces of other races—a phenomenon often referred to as the 'other-race effect'. Using brief presentations of faces, we show a similar effect for the task of discriminating the sex of a face. Specifically, Caucasian observers discriminated male and female Caucasian faces more accurately/efficiently than did Oriental[1] observers, and Oriental observers discriminated male and female Japanese faces more accurately/efficiently than did Caucasian observers. This result indicates that, under suboptimal viewing conditions, the identification of even the most salient of facial characteristics—face sex—is impaired for other-race faces. This finding suggests, also, that the nature and diversity of our experience with faces may affect not only the quality of the face representation for later access by recognition processes, but also the efficiency of a perceptual discrimination process. Intriguingly, too, we found that female observers, for both races tested, were considerably more accurate at the sex classification task than were male observers.

1 Introduction

Human memory for faces is remarkable in some ways, but limited in others. One well-known limiting factor is the difficulty we experience in recognizing faces of other races [cf Shepherd (1981) for a review, and Shapiro and Penrod (1986) for an overview/meta-analysis of the effect in the literature]. By recognition, we mean simply the decision of whether or not we 'know' a face. Associated with this 'other-race' phenomenon is the oft-reported feeling that other-race faces 'all look alike'. The impression that other-race faces are more similar to each other than are own-race faces suggests that we find it difficult to encode the information in individual other-race faces that makes them unique. Indeed, to accurately recognize a face, we must encode something special about the face that distinguishes it from all other faces.

The other-race effect in face recognition is generally thought to result from the vastly different amounts of experience we have with own-race as opposed to other-race faces. Recently, perceptual learning has been proposed as an important mechanism underlying the effects of differential experience on processing own-race versus other-race faces (O'Toole et al 1995). By this account, faces of different races may be thought of as different statistical categories of faces, each varying about its own prototype (O'Toole et al 1991). With experience, we become adept at detecting the features that are most useful for distinguishing among faces of our own race. While paying special attention to these features may result in salient perceptual contrasts for faces of one's own race, it seems to result in much less salient perceptual contrasts for faces of other races.

This perceptual learning account of expertise for same-race faces is not unlike what is generally thought to occur during learning of one's native language (Werker and

[1] We use the British English term 'Oriental' to refer to people from countries in the Far East. The term 'Asian' is more commonly used in American English, but is avoided here since in British English it generally refers to people from India and Pakistan.

Tees 1984; Kuhl et al 1992; Flege et al 1994). With experience, we become adept at detecting the linguistic features most useful for distinguishing among speech sounds in our own language. Unfortunately, these features are not optimally suited for making phonemic contrasts in other languages. Thus, in applying them to the analysis of words in another language, we often experience difficulty in even 'hearing'/perceiving linguistic contrasts that are apparently quite distinct for native speakers of the other language (Werker and Tees 1984). This compelling feeling of perceptual indistinguishability in language seems reminiscent of what we experience when we look at other-race faces.

In addition to our ability tc recognize faces, we are also able to discriminate faces along a number of physical dimensions, including sex, race, and approximate age. In their model of face recognition, Bruce and Young (1986) introduced the term 'visually-derived semantic information' to refer to the features needed to discriminate faces along these dimensions. By comparison to recognition memory tasks, which have supported an other-race effect for faces, tasks involving the discrimination of faces along visually-derived semantic dimensions would seem to be more analogous to those used in the domain of language perception. The major difference between recognition and discrimination concerns the kinds of information required for the tasks. Specifically, to recognize faces we must be able to encode something unique about them. By contrast, to discriminate faces along sex, race, or age dimensions we must be able to perceive the features in individual faces that are common to an entire group of faces, but which also serve to distinguish a face group from alternative group(s) (eg features that all male faces have, but which female faces typically do not have). In other words, these tasks involve a perceptual discrimination between *categories* of faces.

The purpose of the present study was to examine the possibility that 'other-race' factors affect not only the quality of the representation for later access by recognition processes, but also the efficiency of a perceptual discrimination process—specifically, sex discrimination. Face sex categories are psychologically different from ethnic/race categories, because human observers do not typically have more experience with one sex of faces than with the other. While there is reason to think that many of the facial features useful for determining the sex of a face are common to faces of all races (Enlow 1982), these features appear in a facial context/prototype that may differ substantially between different races of faces.[2] We compared the accuracy/efficiency of sex categorization for own-race versus other-race faces. To our knowledge, differences in the efficiency with which a salient facial characteristic such as face sex can be perceived as a function of other-race factors have not been documented, yet would seem to have important implications for applied and theoretical approaches to understanding human memory for faces.

2 Experiment: Classifying own-race and other-race faces by sex

2.1 *Methods*

2.1.1 *Stimuli.* The faces consisted of digitized photographs of 60 Caucasian (30 male, 30 female) and 60 Japanese (30 male, 30 female) young adult faces, without eyeglasses or facial hair. The digitized faces were 152 by 225 (width by height) pixel images, with a resolution of 256 gray levels. All faces were cropped digitally with the aid of Adobe Photoshop™ to eliminate hair length/style cues to face sex. This was done by embedding the face into an oval shape that eliminated completely the external hair shape, and hence also hair length cues to face sex. In some faces, hair hanging onto the forehead in the form of bangs remained. This forehead hair, however, seemed to be present in roughly equal numbers of male and female faces.

[2] Though see Yamaguchi et al (1995) for a discussion of the possibility that there may be some features differentially useful for sex discrimination in different races of faces.

2.1.2 *Observers.* All observers were recruited from the University of Texas at Dallas (UTD) community. Some observers were compensated for their participation with a research credit in a core psychology course at UTD. The Caucasian observers consisted of 10 males and 10 females. Oriental observers ($n = 15$; 9 females and 6 males) comprised Koreans, Chinese, Vietnamese, and a Filipino, varying in the length of US residence from 3 months to 40 years. While clearly it would have been preferable to test only Japanese observers, a previous study (O'Toole et al 1994), in which these same faces were used and observers with similar racial/ethnic backgrounds were tested produced a reliable other-race effect for recognition. This verified that observer membership in the more general category of Oriental was sufficient to produce an other-race effect for Japanese faces.

2.1.3 *Procedure.* Observers read a brief instruction sheet informing them that the purpose of the task was to find out how accurately people could determine the sex of own-race versus other-race faces, when the faces were presented very briefly. The trials proceeded as follows. Individual faces were presented for 75 ms on a computer screen, followed by a 200 ms mask, and then immediately by a visual prompt asking "Male or Female?". The exposure time of 75 ms was chosen, on the basis of pilot data, to yield good, but not perfect, performance. Observers responded "Male" or "Female", without time constraints, using the appropriately labeled key on the computer keyboard. The next trial was initiated when the observer responded.

The faces were blocked by face race, and the order of blocks was counterbalanced across observers within the Caucasian and Oriental observer groups. The order of the faces was randomized individually for each observer.

2.2 *Results*

Performance was assessed as follows. A d' for discriminating male and female faces was computed for each observer on each race of faces. We used d' since we thought it possible that individual observers, or even groups of observers (race or sex), could show a systematic bias to respond "Male" or "Female". For this reason, we also measured the response criterion[3] for individual observers and analyzed these data statistically as well.

2.2.1 *Discrimination.* The d' data were analyzed with a two-factor analysis of variance (ANOVA) with observer race (between-subjects) and face race (within-subjects) used as factors. We found no difference as a function of observer race ($M_{\text{Oriental}} = 2.32$; $M_{\text{Caucasian}} = 2.28$). Sex discrimination performance differed as a function of face race, with Japanese faces classified less accurately than Caucasian faces ($M_{\text{Japanese}} = 2.08$; $M_{\text{Caucasian}} = 2.52$; $F_{1,31} = 19.92$, $p < 0.001$). This result may well be artifactual as more Caucasian than Japanese female faces had noticeably plucked eyebrows, a good social/cosmetic cue to face sex. Interestingly, in a pilot study with these same faces, including the hair, the opposite effect was found—in that case, the sex of the Japanese faces was identified more accurately than the sex of the Caucasian faces. A close look at hairstyles indicated that the Japanese faces had more sexually stereotyped haircuts than did the Caucasians, which was a primary motivation for digitally trimming the hair in the present study.

Additionally, it is perhaps worth noting that there are two other reasonable possibilities for this main effect of face sex. First, all of the observers tested were presently living in the USA, and thus all may have had somewhat more experience with Caucasian than Japanese faces. A second possibility is that the category level of

[3] We used C, a measure of the displacement of the criterion in z-score units, computed as $-0.5(z_{\text{H}} + z_{\text{FA}})$, where the subscripts H and FA refer to hit rate and false alarm rate. With this measure, smaller values imply looser criteria (Snodgrass and Corwin 1988).

'Japanese' may not be exactly comparable to the category level of 'Caucasian'. Rather, Japanese faces are perhaps more appropriately considered as a subcategory of Oriental faces, just as Italian or Swedish faces can be considered subcategories of Caucasian faces. As such, there may have been some differences in the homogeneity of other Caucasian versus Japanese faces due to this category level difference. In any case, while it is certainly possible that these other factors could have contributed to the Caucasian face advantage found, the social/cosmetic factors (trimming the hair) were clearly strong enough to change the direction of the effect.

Of primary interest, however, was the interaction between observer and face race ($F_{1,31} = 6.07$, $p < 0.025$). Figure 1 shows the pattern of means, with the mean performance as a function of face race subtracted. The data show that the Oriental observers performed better than the Caucasian observers on the Japanese faces and that the Caucasian observers performed better than the Oriental observers on the Caucasian faces. This indicates the presence of an 'other-race effect' for classifying faces by sex. For completeness, the original means for the four groups, prior to subtracting out the mean for face race, were as follows: Caucasian observers with Caucasian faces = 2.60; Oriental observers with Caucasian faces = 2.43; Caucasian observers with Japanese faces = 1.97; Oriental observers with Japanese faces = 2.22.

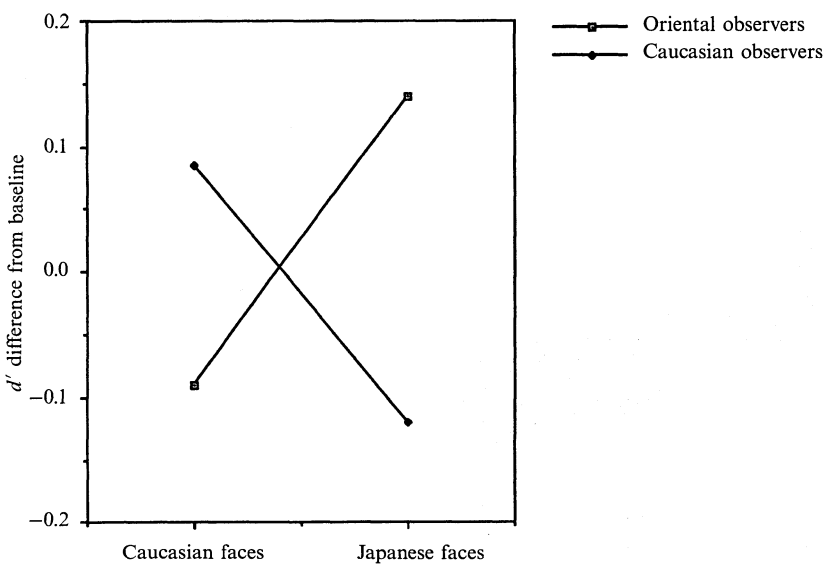

Figure 1. Sex discrimination accuracy as a function of observer and face race. The pattern of means indicates that Caucasian observers discriminated the sex of Caucasian faces more accurately than did Oriental observers, and that Oriental observers discriminated the sex of Japanese faces more accurately than did Caucasian observers.

2.2.2 *Response bias.* We calculated a response criterion for each observer on each race of faces. Hits were defined, arbitrarily, to be the identification of male faces as male, and so false alarms were defined to be the identification of female faces as male. Thus, positive values of the criterion indicate a response bias to identify faces as female, whereas negative values indicate a response bias to identify faces as male. These criterion data were analyzed with a two-factor ANOVA with observer race (between-subjects) and face race (within-subjects) as factors. There was no main effect of observer race or face race, and no interaction between the two. In all the cases the

F values were less than 1. Across all of the faces and both races of observers, however, there was a strong bias to identify faces as male ($M_{\text{Criterion}} = -0.2514$). This criterion was reliably less than zero ($t_{69} = 6.207$, $p < 0.001$).

2.2.3 *Sex-of-observer effects.* While an investigation of the effects of the sex of observer on sex classification performance was not our primary interest, simple informal inspection of the data suggested that it was an important performance variable. As such, we added observer sex to both the criterion and d' ANOVAs. For the criterion ANOVA, no effect of observer sex was found. For the d' ANOVA, however, we found a large difference in sex classification accuracy as a function of this variable, with females ($M_{\text{Female}} = 2.52$) performing more accurately than males ($M_{\text{Male}} = 2.04$), $F_{1,31} = 21.9$, $p < 0.001$. Female observers of both races performed more accurately than did male observers. Sex classification performance (again, with the mean performance as a function of face race subtracted), in ranked order from best to worst, as a function of observer race, observer sex, and face race, shows no exceptions to the better performance of the female over male observers. More formally, observer sex did not interact with either observer race or face race. The pronounced and consistent effect of this variable does not appear to be artifactual in any way. It is worth noting that while differences in the *recognizability* of male versus female faces by male and female observers are minimal and not very reliably found (Shepherd 1981), when effects have been found, they have tended to indicate an advantage for female observers, and also tended to indicate that female observers are particularly accurate with female faces (Shepherd 1981).

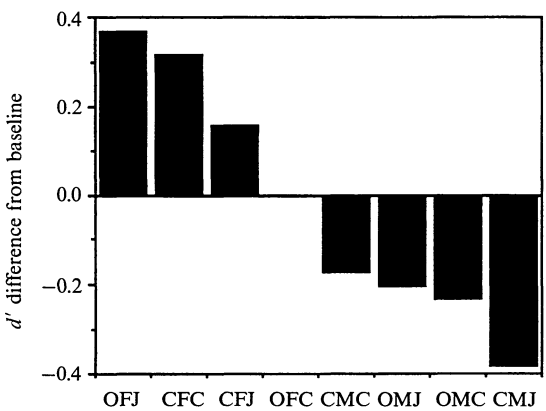

Figure 2. Sex classification performance (with the mean performance as a function of face race subtracted), in ranked order from best to worst, as a function of observer race (C/O), observer sex (F/M), and face race (C/J). C = Caucasian, O = Oriental, J = Japanese, F = female, M = male.

3 Discussion

The present study demonstrated an other-race effect for classifying faces by sex. More specifically, faces of a given race were classified more accurately by members of the same race as opposed to members of another race. This held for both Caucasian and Japanese faces. These results suggest that the nature and diversity of our experience with faces may affect not only the quality of the representation for later access by recognition processes, but also the efficiency of a perceptual discrimination process. The effects of the statistical structure of experience on development is widely evident in both the neural (eg Blakemore and Cooper 1970; Field 1987; Miyashita 1988) and perceptual domains (eg Gregory 1972; Werker and Tees 1984; Kuhl et al 1992). Perceptual effects have been demonstrated perhaps most compellingly in the domain of language,

where phonetic structure differs widely between languages, and where human experience begins early but is frequently limited to a single language. Faces provide a good visual analogy to language, in that facial structure differs widely between races and human experience begins early, but, again, is frequently limited primarily to a single race of faces.

In the broader context of studying the effects of experience on developing brain and perceptual processes conjointly, faces may be a more appropriate stimulus than language to investigate, because animal and human visual abilities are likely to be more comparable than are their acoustic communication abilities. A foundation for this study has been laid in the detailing of neural mechanisms selective for faces/face parts (eg Perrett et al 1992) and in psychological data suggesting the greater importance of developmental exposure to other-race faces (Cross et al 1971; Feinman and Entwistle 1976; Chance et al 1982), over frequency-of-contact in later life (Malpass and Kravitz 1969; Lavrakas et al 1976; Brigham and Barkowitz 1978) in the other-race effect for recognition.

Additionally, the present data raise a number of issues concerning the practical implications of performance with own-race versus other-race faces. Our finding that faces of a given race were classified more accurately by members of their own race than by members of another race should be distinguished from the more generic other-race effect that occurs when observers show better performance for own-race as opposed to other-race faces. In the present case, the strong main effects of face race and observer sex on discrimination performance underlie some interesting exceptions to the generic other-race effect. For example, in absolute terms, Oriental observers performed better on Caucasian faces than on Japanese faces. In the context of the pilot study, for which we found exactly the opposite effect using these same faces with the hair styles intact, the main effect of face race indicates the importance of experience, not only with faces per se, but also with the social and cosmetic factors available for distinguishing male and female faces in everyday life. Presumably, in the present experiment these cues were more informative for the Caucasian faces than for the Japanese faces. Combined, the opposite main effect found in the pilot versus present study, with only a cosmetic change to the faces, indicates rather strongly that, when these social/cosmetic cues are available, observers use them. Separating out the use of these social cues from purely facial cues still remains an important open issue.

A second important performance factor in this study was the sex of the observer. The advantage of female over male observers was robust across both observer and face race variables. The consistency of this finding with previous findings in recognition memory studies (cf Shepherd 1981) suggests that the advantage of female observers with faces is not limited to the context of a recognition task, but may constitute a more general advantage in representing and/or processing faces. While we have no explanation for this effect, the data do provide some hints about aspects of performance that did not differ between male and female observers. In particular, the criterion for identifying a face as male versus female did not differ with the sex of the observer. This is consistent with (though by no means a proof of) the contention that male and female observers did not differ strongly in the strategy they used to classify faces by sex. For example, it might have been possible that the female observers relied on different kinds of features from male observers and that these features were more reliable for determining the sex of the face. As noted, features useful for discriminating male and female faces can be found both in the face itself and in social/cosmetic cues such as plucked eyebrows and hair styles. Given that our Caucasian and Japanese face sets varied in the extent to which these cues may have remained intact in the faces (eg Caucasian female faces had plucked eyebrows), differences in the extent to which male and female observers relied on these different kinds of cues might reasonably have resulted in

criterion effects. That fact that criterion did not differ as a function of observer sex is at least consistent with the idea that male and female observers were using similar features in a similar fashion to make their classification decisions.

In the absence of clear evidence that male and female observers relied on different features to classify faces by sex, it is perhaps worth considering the possibility that the advantage of female observers stems from an ability to extract the features relevant for the task more quickly or more efficiently than male observers. In the present study, exposure times were limited to 75 ms. A task with less time constraints than the task used in the present study might be useful for examining this possibility.

Finally, the existence of an other-race effect for sex classification has some interesting implications for evaluating the credibility of eyewitness descriptions for own-race versus other-race faces. These descriptions often begin with gross categorical information about a face, such as, eg, "It was a Caucasian male, between the ages of ...". Further, the descriptions are often made after encounters that occur under less than optimal viewing conditions. The present data suggest that, under these circumstances, the race of the observer with respect to the race of face being described is potentially one important factor in the accuracy of the description.

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